

AYUSH KUMAR

AI / ML ENGINEER

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PROFESSIONAL SUMMARY

Computer Science undergraduate (B.Tech, expected 2027) who builds end-to-end Machine Learning and Generative AI systems in Python and SQL. Hands-on across the ML lifecycle — supervised learning, NLP, and computer vision with scikit-learn, NumPy, Pandas, and OpenCV — and across modern GenAI: Retrieval-Augmented Generation with vector databases, LLM-powered agentic systems, and prompt engineering with the Gemini API. Comfortable taking a problem from data and EDA through modelling, evaluation, and cloud deployment, and communicating results to stakeholders. Strong DSA foundation with 300+ problems solved in Java.

EDUCATION

B.Tech, Computer Science and Engineering

VIT Bhopal University, India | CGPA: 8.56 / 10

Sep 2023 – May 2027 (Expected)

Senior Secondary (Class XII), CBSE — 92.8%

Sachdeva Millennius School, India

2022

Secondary (Class X), CBSE — 90.5%

Sachdeva Millennius School, India

2020

PROJECTS

VoxAide AI — Agentic Voice Assistant (RAG + Multi-Tenant) / Personal Project

[voxaide.web.app](#)

React, Python, Flask, ChromaDB, Gemini API, text-embedding-004, Speech-to-Text, Twilio, Firebase

- Built an **agentic RAG system** that answers live customer calls: a channel-agnostic agent orchestrator retrieves company knowledge, reasons with Gemini, and executes actions via **function calling**.
- Implemented the **RAG pipeline** end to end — document chunking, Gemini embeddings, and semantic similarity search over a **ChromaDB vector store** with strict per-company (multi-tenant) isolation so one client's data never reaches another.
- Designed an extensible agent **tool registry** (appointment booking, human escalation) with structured JSON schemas, and added anti-hallucination guardrails that intercept and correct fabricated actions before they reach the caller.
- Integrated Twilio voice with speech-to-text and text-to-speech so the agent holds real phone conversations, and deployed the frontend and backend as independent services for zero-downtime updates.

EchoGrow — Generative AI Business Growth Platform / Personal Project

[echogrow-three.vercel.app](#)

Next.js, PostgreSQL, Prisma ORM, Google Gemini API, Tailwind CSS, Vercel

- Developed the **AI campaign-generation endpoint** on the Gemini API with a strict JSON response schema and an automatic fallback model, returning reliable, structured marketing storyboards to the UI.
- Applied **prompt engineering** to convert raw business inputs into consistent, structured GenAI outputs, tuning prompts and schemas to eliminate malformed responses.
- Designed a 3-table PostgreSQL schema (leads, subscribers, campaigns) with Prisma ORM and built the backend as typed server actions and an API route with validation on every submission.
- Built a 10-route app with 13 reusable React components and a lead-management dashboard, secured admin access with an HMAC-signed session cookie, wrote 30 automated end-to-end tests, and deployed on Vercel.

E-commerce Recommendation System — NLP & Embeddings / Machine Learning Project

[GitHub](#)

Python, NumPy, Pandas, scikit-learn, Word2Vec, Jupyter

- Built a **content-based recommendation engine** that suggests semantically similar apparel products directly from their text descriptions.
- Engineered text-similarity features with **TF-IDF** and **Word2Vec** word embeddings, computing semantic similarity between products — the same embedding-and-retrieval principle that powers vector search in RAG.
- Compared multiple text-representation approaches and selected embeddings for the best relevance-versus-coverage trade-off in the recommendations.
- Engineered and optimized the NLP pipeline in Python (NumPy, Pandas, scikit-learn) to serve recommendations at **sub-100ms latency** for scalable performance.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java, HTML5, CSS3

Machine Learning: Regression, KNN, Clustering; scikit-learn, NumPy, Pandas, OpenCV

Generative & Agentic AI: RAG, Vector Databases (ChromaDB), Embeddings, LLMs, Prompt Engineering, Gemini API

Backend & APIs: REST APIs, FastAPI, Flask

Frontend: React.js, Next.js, Tailwind CSS

Databases: PostgreSQL, MySQL, Firebase Firestore

Cloud & Deployment: Google Cloud, Firebase (Auth, Firestore, Hosting), Vercel, Render

Testing & Tools: Automated end-to-end testing, Git, GitHub, VS Code, Jupyter, Twilio API

CERTIFICATIONS

- Advanced RAG with Vector Databases and Retrievers — Coursera
- Build Multimodal Generative AI Applications — Coursera
- Develop Generative AI Applications: Get Started — Coursera

ACHIEVEMENTS

- Solved 300+ algorithmic problems in Java across LeetCode, CodeChef, HackerRank, InterviewBit, and SPOJ.
- Selected for the Ericsson Edge Academia Program among the top 700 candidates worldwide.